

THE UNIVERSITY OF NEW SOUTH WALES
SCHOOL OF PHYSICS

MIDSESSION EXAMINATION

PHYS3114 – Electrodynamics
Term 3, 2023

1. Time allowed – 50 minutes
2. Total number of questions – 2
3. Total marks available – 30 (pro rata to 15% of course assessment)
4. Answer ALL questions
5. PARTS ARE NOT OF EQUAL VALUE.
Marks available for each part of the questions are shown in the examination paper.
6. Answer all questions in a single book.
7. University-approved calculators may be used.
8. All answers must be written in ink. Except where they are expressly required, pencils may only be used for drawing, sketching or graphical work.
9. This paper may be retained by the candidate.

PHYS3114 — Definitions and Formulae

$$F(s) = \int_{-\infty}^{+\infty} f(x) e^{-2\pi i x s} dx \quad (1)$$

$$\mathcal{F}\{f(x)\} = 2 \int_0^\infty f_e(x) \cos(2\pi s x) dx - 2i \int_0^\infty f_o(x) \sin(2\pi s x) dx \quad (2)$$

$$V(x', y') = C_1 C_2 \lambda^2 f_o^2 \iint_{\text{bfp}} F\left(\frac{\zeta}{\lambda f_o}, \frac{\eta}{\lambda f_o}\right) e^{-i2\pi(\frac{f_g}{D} x' \frac{\zeta}{\lambda f_o} + \frac{f_o}{D} y' \frac{\eta}{\lambda f_o})} d\left(\frac{\zeta}{\lambda f_o}\right) d\left(\frac{\eta}{\lambda f_o}\right) \quad (3)$$

$$\gamma(P_1, P_2) = \frac{\iint_{\text{source}} I(\zeta, \eta) e^{\frac{-2\pi i}{\lambda}(p\zeta + q\eta)} d\zeta d\eta}{\iint_{\text{source}} I(\zeta, \eta) d\zeta d\eta} \quad (4)$$

$$F(\rho) = 2\pi \int_0^\infty f(r) J_0(2\pi \rho r) r dr \quad (5)$$

$$\frac{m\lambda}{2} = \frac{r_m^2}{2} \left(\frac{1}{r_0} + \frac{1}{b_0} \right) \quad (6)$$

$$\frac{d^2 P}{dt^2} + \gamma \frac{dP}{dt} + \omega_0^2 P = \frac{e^2 N}{m} \left(E + \frac{\nu P}{\epsilon_0} \right). \quad (7)$$

$$\hat{K} - 1 = \frac{\omega_p^2(\omega_{0'}^2 - \omega^2)}{(\omega_{0'}^2 - \omega^2)^2 + (\gamma\omega)^2} + i \frac{\omega_p^2 \omega \gamma}{(\omega_{0'}^2 - \omega^2)^2 + (\gamma\omega)^2} \quad (8)$$

$$\mathbf{T} = \begin{pmatrix} 1 & 0 \\ \frac{t}{n} & 1 \end{pmatrix} \quad (9)$$

$$\mathbf{R} = \begin{pmatrix} 1 & \frac{n_i - n_t}{R} \\ 0 & 1 \end{pmatrix} \quad (10)$$

Theorem	$f(x)$	$F(s)$
Similarity	$f(ax)$	$\frac{1}{ a }F\left(\frac{s}{a}\right)$
Linearity	$\alpha \cdot f(x) + \beta \cdot g(x)$	$\alpha \cdot F(s) + \beta \cdot G(s)$
Shift	$f(x - a)$	$e^{-2\pi ias}F(s)$
Convolution	$f(x) * g(x)$	$F(s)G(s)$
Derivative	$\frac{d}{dx}(f(x))$	$2\pi isF(s)$
Rayleigh	$\int_{-\infty}^{+\infty} f(x) ^2 dx$	$\int_{-\infty}^{+\infty} F(s) ^2 ds$

Table 1: Basic theorems of Fourier theory

Question 1 (15 marks)

- (a) Solve the following integral directly,

$$f * g(x) = \int_{-\infty}^{\infty} f(u)g(x-u) du \quad (11)$$

for the convolution of the hat function $f(x) = \prod_{-\frac{1}{2}+\frac{1}{2}}(x)$ with the shifted Dirac delta function $g(x) = \delta(x-a)$. [3 marks]

A phase grating is formed when light passes through a transparent film where the refractive index of the film is varied periodically as a function of position. Such an object can be treated as a diffracting aperture and can be modelled mathematically using a function of the form:

$$f(x) = e^{i\varphi(x)} \quad (12)$$

Consider a phase grating that varies sinusoidally with position with a period of d , and total width of Λ . Here we consider the case where Λ is large compared with d . The phase grating is weak, such that the amplitude of the phase modulation, a , is much less than 1. The aperture is illuminated with monochromatic light of wavelength, λ , from a distant source and the corresponding diffraction pattern is observed under far-field conditions.

- (b) Write down an expression for the function that describes this aperture. [3 marks]
- (c) Determine the diffracted field from this aperture in terms of diffraction angles. Hint: Apply Euler's formula and then use the small angle approximation. [6 marks]
- (d) Make a plot of the corresponding diffracted intensity as a function of diffraction angle if $\Lambda = 100d$ and $d = 10\lambda$. Marking any critical features on the plot. [3 marks]

Question 2 (15 marks)

- (a) Using the matrix method for geometric optics, and providing an explanation of any relevant approximations used, confirm the lens maker's equation for the thin spherical lens:

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad (13)$$

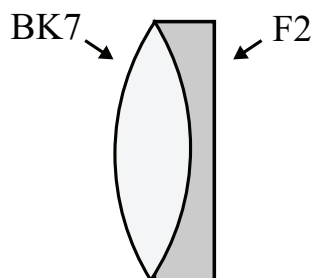
[5 marks]

- (b) Consider a bi-convex thin lens that is composed of BK7 glass and has respective radii of curvature, $|R_1| = |R_2| = 10$ cm. Determine the variation in the focal length of this lens between 400 nm and 700 nm given the following refractive index values in the table below: [2 marks]

λ	n_{BK7}	n_{F2}
400 nm	1.53	1.65
700 nm	1.51	1.61

- (c) Now consider a thin compound lens made from combining the bi-convex lens above with a second plano-concave lens made of F2 flint glass with $|R_1| = 10$ cm and $|R_2| = \infty$, as depicted in the figure. Derive an expression for the series of matrix multiplications that would represent propagation through this lens system. Hint: you needn't consider the thickness of either lens element.

[4 marks]



- (d) Use the above result to derive an expression for the focal length of this compound lens and compare the wavelength dependence of the focal length with the single element lens. Comment on the significance of the result. [4 marks]